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The Editors
Journal of Hazardous Materials

Dear Editors,

We submit the research article *"A testing-effort artefact in the most-sensitive aquatic endpoint: structure-based PFAS screening recovers predictive validity but still cannot support clearance"* for consideration in *Journal of Hazardous Materials*.

Is the studied subject an environmental contaminant? The work concerns per- and polyfluoroalkyl substances, analysed through 4,744 distinct measured freshwater aquatic toxicity records for 87 parent structures drawn from the U.S. EPA ECOTOX Knowledgebase, and applied to 17,619 untested structures from the EPA PFASSTRUCTV6 inventory. The broader diagnostic uses 106,095 freshwater acute records across 4,757 chemicals.

Is the study conducted under environmentally relevant conditions? We answer this directly rather than by reference to the data being measured. PFAS occur in surface water at ng/L to low µg/L; the EU environmental quality standard for PFOS is 0.65 ng/L and the US maximum contaminant level for PFOA and PFOS is 4 ng/L. The effect concentrations modelled here, and the 0.1 to 1 mg/L decision boundaries applied to them, sit several orders of magnitude above that range — as every hazard term does, since a predicted no-effect concentration is an effect concentration divided by an assessment factor. The environmental relevance of this work lies in what happens to that division: an assessment factor is a multiplicative constant and therefore leaves a 4.9 $\log_{10}$ unit effort gradient in the hazard term completely intact. A criterion derived at nanograms per litre from a minimum-based hazard term inherits the whole of it. We make this argument explicitly in §3.7 and mark the occurrence range on Figure 2c so that a reader can see where real concentrations sit relative to the modelled ones. The required 100-word environmental implication statement accompanies this submission as a separate document.

A note on provenance. This manuscript is a complete rebuild of work previously declined by another journal, and we have treated the referees' criticism of our data curation as correct. The per-substance table has been regenerated so that structure standardisation precedes rather than follows endpoint summarisation; exact duplicate records are now removed before testing effort is counted; the structure-class analysis has been recomputed on the present extraction, which removes a positive result we previously reported; and a claim that representation choice does not matter has been withdrawn as contradicted by our own deposited results. Each correction is stated in the text where it applies rather than only in the corrected number. The central finding is unchanged and slightly stronger after correction.

The contribution. Regulatory hazard screening conventionally summarises a substance by the lowest effect concentration reported for it. We show that this makes the response variable depend on how often the substance has been tested: across PFAS the protective minimum correlates with

record count at $\rho = -0.75$, and subsampling to equal depth, compared against the correct eligible-subset baseline, identifies 40–48% of that as a pure extreme-order-statistic artefact. Because a structure-based model cannot predict testing history, this confound is by itself sufficient to produce the external validation failures reported in the PFAS screening literature. Holding a published model fixed and changing only how the same records are summarised raises rank transfer on the same 58 structures by $\Delta\rho = +0.32$ (95% CI $[+0.08, +0.57]$). Rebuilding training and external sets consistently on an effort-stable endpoint restores external validity and narrows the prediction interval — and yet no untested structure clears at a stringency-matched threshold.

We believe this is useful to the journal's readership in two ways. It identifies a measurable defect in a widely used data-preparation convention, with a diagnostic that any screening study can compute in one line. And it corrects the interpretation of a negative result that is becoming established: structure-based models are not failing on PFAS because they cannot represent fluorinated chemistry, but substantially because they are being asked to predict a quantity that encodes testing effort.

Prior submission. An earlier and materially different version of this work, based on a curated subset of the aquatic record and without the testing-effort analysis, was reviewed and declined by *Environmental Science & Technology* (es-2026-093898). The present manuscript uses a different data source, reports a different central finding, and reaches its conclusion by a different mechanism; we mention the prior review in the interest of transparency and can supply those reviews on request.

All extraction and analysis scripts, structure-locked tables and inventory-level predictions are openly available and tagged as an immutable release. The manuscript is original, is not under consideration elsewhere, and is approved by all authors. The authors declare no competing financial interest.

Sincerely,

Ku Kang, on behalf of all authors